Olin — Experiment Report 01

Einstein Radius Estimation Using Convolutional Neural Networks

1 Objective and Experimental Scope

The task is a single-target regression problem: given a final noisy synthetic lensing image, predict its generating Einstein radius, θ_E. In Dataset V0, θ_E varies over the range 1.6–2.8 arcsec while the other lens, source, and observation configuration parameters are held fixed. This controlled design isolates the initial one-parameter inverse-inference task.

2 Dataset and Forward-Model Configuration

Table 1: Dataset V0 configuration

Configuration	Value
Number of synthetic images	10,000
Image dimensions	160×160 pixels
Pixel scale	0.05 arcsec/pixel
Field of view	8×8 arcsec
Einstein radius range	1.6–2.8 arcsec
Lens model	Singular Isothermal Ellipsoid (SIE) + external shear
Source model	Elliptical S
'ersic profile	
PSF	Gaussian; FWHM = 0.4 arcsec
Photon scale	1,000
Read noise	1.0
Train / validation / test split	8,000 / 1,000 / 1,000

The CNN receives only the final noisy image. The target label is the known Einstein radius used to generate that image. Dataset normalization uses statistics calculated from the training split and then applies those same statistics to the validation and test splits.

3 CNN Architecture

CNN_0 and CNN_1 share the same architecture:

Table 2: CNN architecture

Layer	Configuration
Input	$160 \times 160 \times 1$
Convolution 1	16 filters, 3×3, ReLU
Max pooling 1	2×2
Convolution 2	32 filters, 3×3, ReLU
Max pooling 2	2×2
Flatten	Flatten feature maps
Dense	64 units, ReLU
Output	1 unit, linear activation

The model contains approximately 2.96 million trainable parameters. Both runs used Adam, mean squared error (MSE) loss, mean absolute error (MAE) as the reported metric, batch size 32, and random seed 42. CNN_1 was freshly initialized; it was not fine-tuned from CNN_0.

4 Training Experiments

Table 3: Training-procedure comparison

Setting	CNN_0	CNN_1
Maximum epochs	10	30
Early stopping	Not used	Monitor validation MAE; patience 5; restore best weights
Learning-rate reduction	Not used	Factor 0.5; patience 2; minimum learning rate 10^{-6}
Checkpointing	No best-validation checkpoint selection	Save checkpoint with lowest validation MAE
Epochs completed	10	14
Best validation MAE	Not recorded for this comparison	0.01848944 arcsec (epoch 9)

CNN_1 used validation MAE to select the best checkpoint. The maximum epoch count was 30, but training stopped after 14 epochs under the configured early-stopping rule. The best checkpoint, rather than the final epoch weights, was used for test-set evaluation.

5 Comparative Test Results

Table 4: Held-out test-set performance

Metric	CNN_0	CNN_1
Test MSE	0.01388483	0.00057045
Test MAE (arcsec)	0.11139602	0.01924538

CNN_1 reduced test MAE by approximately 82.7% relative to CNN_0. Since the architecture, dataset, optimizer, loss, and batch size were retained, this comparison is consistent with a benefit from the revised training procedure. However, the callbacks were introduced together, so this experiment does not identify the independent contribution of each callback.

6 CNN_1 Test-Set Error Analysis

The following statistics were calculated over the 1,000 held-out test samples.

Table 5: CNN_1 test-set error statistics

Metric	Result
Mean absolute error	0.0192453837 arcsec
Median absolute error	0.0167946167 arcsec
90th percentile absolute error	0.039314 arcsec
95th percentile absolute error	0.046847 arcsec
Maximum absolute error	0.070672 arcsec
Mean signed error	−0.003671 arcsec

The negative mean signed error indicates a small aggregate tendency toward underestimation. The mean absolute error exceeds the median, consistent with a right-tailed distribution of absolute errors.

7 Diagnostic Visualizations

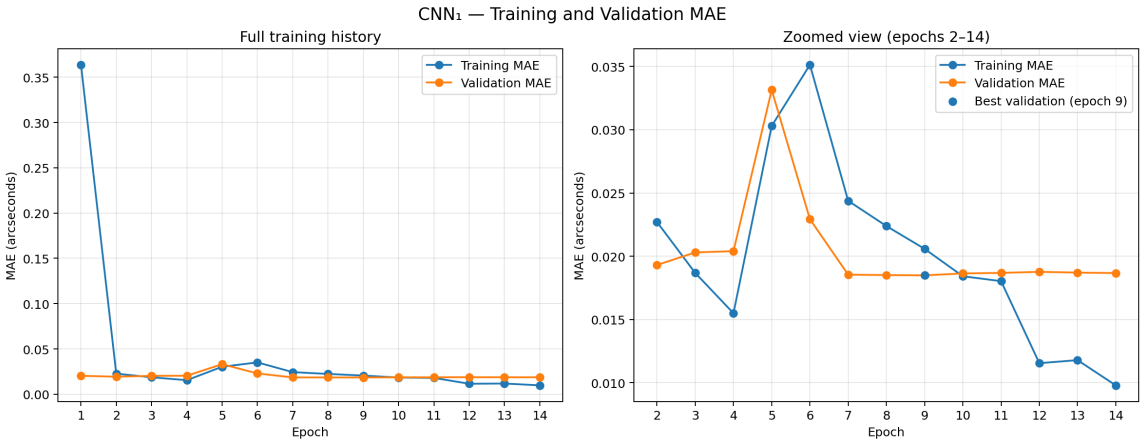


Figure 1: CNN_1 training and validation MAE over 14 epochs. The full view includes the unusually high first-epoch training MAE; the zoomed view makes epochs 2–14 easier to inspect. The lowest recorded validation MAE occurred at epoch 9.

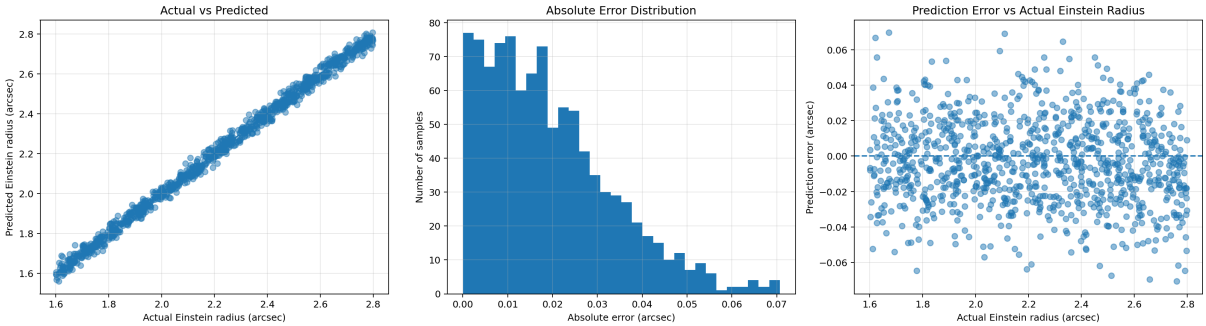


Figure 2: CNN_1 test-set prediction diagnostics for 1,000 samples. Left: actual versus predicted Einstein radius, with the identity line representing perfect predictions. Centre: absolute-error histogram. Right: signed prediction error versus true Einstein radius, with a zero-error reference.

The actual-versus-predicted plot shows close visual agreement with the identity line over the test range. The residual plot does not reveal an obvious strong trend by visual inspection, but visual inspection alone is not a formal test of residual independence or uniform error across the range.

8 Interpretation and Limitations

The evaluation is limited to a held-out split drawn from Dataset V0. Since other simulation parameters were fixed, the experiment does not establish robustness to changes in lens ellipticity, external shear, source morphology, or observing conditions. It also does not establish accuracy on real astronomical observations.

The results should therefore be interpreted as a controlled proof of concept for estimating one lens parameter within the Dataset V0 simulation configuration, not as evidence that the model is ready for scientific inference on arbitrary lensing observations.

9 Conclusion and Next Validation

Under Dataset V0 conditions, CNN_1 achieved substantially lower test error than CNN_0 for estimating the Einstein radius from synthetic images. This provides an initial controlled baseline for Olin's inverse-inference pipeline.

The next step is to quantify error across Einstein-radius intervals, followed by controlled tests that vary source, lens, and observation parameters. Claims of real-data generalization should wait until a compatible external validation study has been completed.

10 Reproducibility Artifacts

The experiment is associated with the following scripts and outputs:

- `train_cnn_0.py` and `train_cnn_1.py`: training scripts.
- `analyze_cnn_1_errors.py`: test-set prediction and error analysis.
- `plot_cnn_1_evaluation.py`: test diagnostic plotting script.
- CNN_1 best checkpoint: `cnn_1_best.keras`.
- Training history and metrics: `history.json` and `metrics.json`.
- Test predictions: `predictions.npy`.

- Figures: `cnn_1_learning_curves.png` and `test_error_analysis.png`.

Dataset arrays and model checkpoints are maintained separately from this report. Verify the presence and exact filenames of the artifacts in the project directory before archiving or publishing the experiment.

Report note: This document records the experiment details and results shared during the project discussion. It is an experiment record, not an independent audit of the complete local project.